Midterm examination · Sample solutions

Q1 · Short answers (20 points)

(a) Step size (10 points)

No. A smaller step can amplify floating-point cancellation or roundoff, so the observed improvement from 0.1 to 0.01 to 0.001 does not establish that reducing the step indefinitely will improve accuracy. One sound reason is sufficient.

(b) Newton–Raphson (10 points)

Yes, under the mathematical Newton–Raphson update:

$$x_1 = x_0 - \frac{ax_0 + b}{a} = -\frac{b}{a}.$$

Since $a \neq 0$, the derivative is nonzero and this is the root for every initial guess. If the initial guess is already the root, no update is needed. A reasoned qualification about finite-precision arithmetic is also acceptable.

Q2 · Peng–Robinson (40 points)

(a) Residual (10 points)

Move the specified pressure to the other side, so that the difference between calculated and specified pressure is zero:

$$F(V; T, P) = \frac{RT}{V - b} - \frac{a(T)}{V^2 + 2bV - b^2} - P = 0.$$

The unknown is molar volume V, with T and P held fixed. The opposite residual sign is equally valid. V is in L/mol and F in bar.

$$\kappa = 0.70647745296, \qquad b = 0.026667898117 \text{ L mol}^{-1}.$$

(b) Numerical implementation (10 points)

```
import numpy as np
from scipy.optimize import root_scalar

Tc, Pc, omega, R = 304.1282, 73.7703, 0.22394, 0.0831446
kappa = 0.37464 + 1.54226 * omega - 0.26992 * omega**2
b = 0.07780 * R * Tc / Pc

def a(T):
    return 0.45724 * R**2 * Tc**2 / Pc * (
        1 + kappa * (1 - np.sqrt(T / Tc))
    )**2

def F(V, T, P):
    return R*T/(V-b) - a(T)/(V**2 + 2*b*V - b**2) - P

print(F(1.0, 400.0, 1.0))
```

At 400 K, $a(T) = 3.183575375$ L^2 bar mol^{-2}, and the sanity check gives $F(1; 400, 1) = 30.14464033$ bar. An opposite-sign residual gives the negative of this value.

(c) Root finding (20 points)

Bisection is one suitable method:

```
brackets = [(0.045, 0.065), (0.10, 0.14), (0.35, 0.45)]
for bracket in brackets:
    solution = root_scalar(
        F, args=(280.0, 40.0), bracket=bracket,
        method="bisect", xtol=1e-12,
    )
    print(solution.converged, solution.root, F(solution.root, 280, 40))
```

The endpoint residuals are approximately $(100.899, -33.345)$, $(-9.295, 6.321)$, and $(2.148, -3.578)$ bar. Each interval lies above b, contains no pole, and brackets a root. The roots are

$$\boxed{0.05189863909, \quad 0.11815837363, \quad 0.38528728917 \text{ L mol}^{-1}}.$$

All three solves converge, with residual magnitudes below 4×10^{-9} bar. Independently multiplying the equation by its denominators gives the cubic

$$PV^3 + (Pb - RT)V^2 + (a - 3Pb^2 - 2RTb)V + Pb^3 + RTb^2 - ab = 0,$$

whose three roots agree with the bisection results.

Q3 · Mass spectrometry (40 points)

(a) Formulation (10 points)

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & 3 & 2 \\ 8 & 2 & 4 & 2 \\ 0 & 7 & 2 & 9 \\ 0 & 2 & 9 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_A \\ x_B \\ x_C \\ x_D \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 21 \\ 34 \\ 29 \\ 31 \end{bmatrix}.$$

```
import numpy as np

A = np.array([[4, 1, 3, 2],
              [8, 2, 4, 2],
              [0, 7, 2, 9],
              [0, 2, 9, 0]], dtype=float)
b = np.array([21, 34, 29, 31], dtype=float)
```

(b) Amounts and check (15 points)

```
x = np.linalg.solve(A, b)
calculated_intensities = A @ x
print(x)
print(calculated_intensities, np.allclose(calculated_intensities, b))
```

$$\boxed{\mathbf{x} = (2,\ 2,\ 3,\ 1)^T}, \qquad \mathbf{Ax} = (21,\ 34,\ 29,\ 31)^T.$$

For hand elimination, $R_2 \leftarrow R_2 - 2R_1$ gives the augmented row $(0, 0, -2, -2 \mid -8)$. The second pivot is zero, so swap rows 2 and 3 before proceeding. The matrix is nonsingular.

(c) Measurement correction (10 points)

```
b_true = np.array([22, 34, 29, 31], dtype=float)
x_true = np.linalg.solve(A, b_true)
e_rel = np.abs(x - x_true) / x_true
print(x_true)
print(e_rel)
print(1 / 22)  # sensor relative error
```

$$\boxed{\mathbf{x}_{\text{true}} = \begin{bmatrix} 1.95454545 \\ 0.94805195 \\ 3.23376623 \\ 1.76623377 \end{bmatrix}}.$$

For each compound i, use $|x_i - x_{i,\text{true}}|/x_{i,\text{true}}$.

Compound	Absolute relative error
A	0.023256
B	1.109589
C	0.072289
D	0.433824

Multiplying by 100% gives 2.326%, 110.959%, 7.229%, and 43.382%, respectively.

The sensor's relative error is $1/22 \times 100\% = 4.545\%$.

(d) Error amplification (5 points)

Conditioning describes the sensitivity of the solution to changes in the measurements. Here, a small error in one intensity produces much larger relative errors in some calculated amounts because the four contributions must change together while preserving the other intensities.

For reference, $\kappa_2(\mathbf{A}) \approx 21.24$, from `np.linalg.cond(A)`. A condition-number bound uses vector norms, so it does not directly bound each component's percentage error. Pivoting addresses the zero pivot in elimination but does not remove the system's sensitivity to measurement error.